

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

April/May 2012

EC301 Electromagnetic Theory

Date: 30.04.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) By illustrating examples, define Scalar and Vector Quantities [02]
(b) With help of neat sketches, perform conversion of Cylindrical coordinate systems to Cartesian coordinate Systems [05]
- Q - 2 (a) Explain cross product using advancement of right-handed screw concept. [02]
(b) Derive equation for Electric Field Intensity [05]
(c) Describe Electric flux density D , in brief. Derive equation of Gauss Law. [07]

OR

- Q - 2 (a) Describe Vector Field in brief. [02]
(b) Illustrating diagram, derive equation for Line Integral for work done in moving point charge Q . [05]
(c) Given the potential field, $V = 2x^2y - 5z$ and a point $P(-4, 3, 6)$. At point P calculate: [07]
(i) Potential V (ii) Electric Field Intensity E , (iii) The direction of E (iv) Electric flux density D .
- Q - 3 (a) Mention Maxwell's equation in Integral Form. [02]
(b) Let a filamentary current of 5 mA be directed from infinity to the origin on the positive z axis and then back out to infinity on the positive x axis. Find H at $P(0, 1, 0)$. [05]
(c) Explain Biot-Savart law with necessary diagram and equations. [07]

OR

- Q - 3 (a) Mention Maxwell's equation in Point Form. [02]
(b) Let $G = 15ra_r$, (i) Determine $\oint G \cdot dL$ for the circular path $r = 5$, $\theta = 25^\circ$, $0 \leq \phi \leq 2\pi$. [05]
Evaluate $\oint (\nabla \times G) \cdot dS$ over the spherical cap $r = 5$, $0 \leq \theta \leq 25^\circ$, $0 \leq \phi \leq 2\pi$.
(c) Explain Ampere's circuital law with help of necessary diagram and equations. [07]

SECTION – II

- Q - 4 (a) Give significance and equation of Retarded Potential. [02]
(b) What is displacement current? Derive equation for displacement current [05]
- Q - 5 (a) Obtain Poisson's and Laplace equations. [07]
(b) Explain Force and Torque on Closed Circuit [07]

OR

- Q - 5 (a) Derive Magnetic Boundary Conditions for two different magnetic materials [07]
(b) What is Ferromagnetic Material? With help of necessary equations, explain magnetization and Permeability [07]
- Q - 6 (a) Explain wave propagation in dielectrics to derive equation of intrinsic impedance η . [07]
(b) Derive equation of Poynting Vector and give significance of Poynting Vector. [07]

OR

- Q - 6 (a) Illustrating neat diagrams explain standing wave ratio. [07]
(b) Explain and derive equation for Wave Polarization. [07]
